# Ecology (2026)

Study online at https://quizlet.com/_ffqdq7

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| ecology | scientific study of all the relationships between organisms and their environment |
| ecosystem | biological community and all the nonliving factors that affect it |
| atmosphere | blanket of gases surrounding Earth |
| biosphere | all of Earth's organisms and the environments in which they live |
| population | group of organisms of the same species that occupy the same geographic place at the same time |
| biological community | all the interacting populations of different species that live in the same geographic location at the same time |
| biotic factor | any living factor in an organism's environment |
| abiotic factor | any nonliving factor in an organism's environment; e.g. soil, water, temperature, and light |
| limiting factor | biotic or abiotic factor that restricts the number, distribution, or reproduction of a population within a community |
| tolerance | organism's ability to survive biotic and abiotic factors |
| habitat | physical area in which an organism lives |
| niche | role, or position, of an organism in its environment |
| competition | occurs when more than one organism uses a resource at the same time; e.g. food, water, space, or mates |
| predation | act of one organism hunting and feeding on another organism |
| symbiosis/symbiotic relationship | relationship in which two species live closely together |
| mutualism | symbiotic relationship in which both species benefit from the relationship |

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| commensalism | symbiotic relationship in which one organism benefits and the other is neither helped nor harmed |
| parasitism | symbiotic relationship in which one organism benefits and the other organism is harmed or dies |
| autotroph/producer | organism that captures energy from sunlight to produce its own food; provides the foundation of the food supply for all other organisms |
| heterotroph/consumer | organism that cannot make its own food and gets its nutrients and energy requirements by feeding on other organisms |
| herbivore | heterotroph that consumes (eats) only plants |
| carnivore | heterotroph that consumes (eats) other heterotrophs |
| omnivore | heterotroph that consumes (eats) both plants and animals |
| detritivore | heterotroph that consumes (eats) dead organic matter accelerating the decomposition process; worms, crabs, other bugs |
| decomposer | breaks down dead organisms by releasing digestive enzymes and absorbing nutrients externally; mostly bacteria and fungi |
| food chain | simplified model that shows a single path for energy flow through an ecosystem |
| trophic level | each step in a food chain or food web |
| food web | model that shows many interconnected food chains and pathways in which energy and matter flow through an ecosystem |
| biomass | total mass of living matter at each trophic level |
| biodiversity | number of different species living in a specific area |

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| extinction | the disappearance of a species when the last of its members dies |
| endangered species | species that have a very high risk of extinction in the near feature |
| deforestation | the removal of all trees from a wooded area; a major cause of habitat loss |
| invasive species | introduced species that significantly modify or disrupt a habitat |
| natural resource | resources provided by Earth, including air, water, land, all living organisms, nutrients, rocks and minerals |
| climate change | the change in average temperatures and variations in weather around the world |
| eutrophication | process by which lakes become rich in nutrients from the surrounding watershed, resulting in a change in the kinds of organisms in the lake |
| biomagnification | increasing concentration of toxic substances, such as DDT, in organisms as trophic levels increase in food chains or food webs |
| ecosystem diversity | the variety of ecosystems in the biosphere |
| species diversity | in a biological community, the number and abundance of different species |
| hydrosphere | all the water in Earth's oceans, lakes, seas, rivers, and glaciers plus all the water in the atmosphere |
| biome | large group of ecosystems that share the same climate and have similar types of communities |
| detritus | fragments of dead matter in an ecosystem |

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| carbon sink | natural or artificial systems that absorb and store more carbon dioxide from the atmosphere than they release; oceans, forests, and soil |
| nitrogen fixation | process in which nitrogen gas is captured and converted into a form plants can use |
| nitrification | the chemical process that turns ammonium, NH4+, into the nitrogen-oxygen compounds, NO2- and NO3- |
| denitrification | process in which fixed nitrogen compounds are converted back into nitrogen gas and returned to the atmosphere |
| stability | when something stays mostly the same over time |
| acidification | the ongoing decrease in the pH of the Earth's oceans, primarily caused by absorbing CO2 from the atmosphere |
| atmospheric nitrogen | nitrogen that is not usable by animals and plants, part of the nitrogen cycle |
| carbon cycle | the movement of carbon from the nonliving environment into living things and back |
| ecological pyramid | diagram that shows the relative amounts of energy or matter within each trophic level in a food chain or food web |
| consumer | an organism that obtains energy by feeding on other organisms |
| energy pyramid | a diagram that shows the amount of energy that moves from one feeding level to another in a food web or food chain |
| impact | an effect or result |
| nitrogen cycle | the transfer of nitrogen from the atmosphere to the soil, to living organisms, and back to the atmosphere |

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| organism | an individual living thing |
| primary consumer | an organism that eats producers; herbivore |
| secondary consumer | an organism that eats primary consumers; a carnivore that eats an herbivore |
| tertiary consumer | an organism that eats secondary consumers; carnivore |